

Lab Assessment Rubrics of Digital System Design (CPE-344)										
Lab Number			10							
Lab Title			Implementation of Different type of advance delay model in varilog HDL and FPGA board							
Submitted by										
Names			Registration Number							
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Rubrics name & number							Marks			
							In-Lab	Post-Lab		
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and them using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

Lab: 10

1. Objectives:

The objective of this lab is to get familiar with

- Designing of IIR filter.
- Implementation of IIR filters in Verilog.

2. Lab Tasks:

2.1. In-Lab Task#01:

Implement the following equation of IIR

$$y[n] = 0.5 y[n-1] + x[n]$$

Solution:

- **Verilog Code**

```
module example1(input signed [15:0] x, input clk, input rst_n, output reg signed
[31:0] y);

reg signed [31:0] y_reg;

always @(*) // combinational logic block

y = (y_reg>>>1) + x;
always @(posedge clk or negedge rst_n) // sequential logic block
begin
if (!rst_n)
y_reg <= 0;
else
y_reg <= y;
end
endmodule
```

- **Verilog Test Bench**

```
module example1TestBench;
reg [15:0] x;
reg clk;
reg rst_n;
reg i;

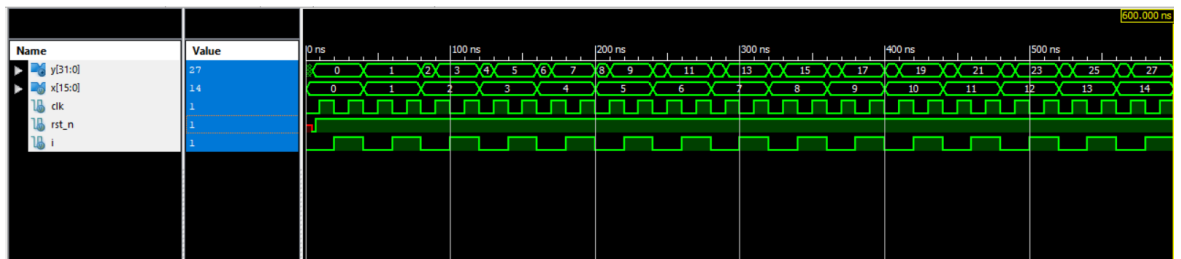
wire [31:0] y;
example1 uut (
.x(x),
.clk(clk),
.rst_n(rst_n),
.y(y)
);
initial begin
clk =0;
#5 rst_n =0; // resetting register before first posedge clk
#2 rst_n= 1;
```

```

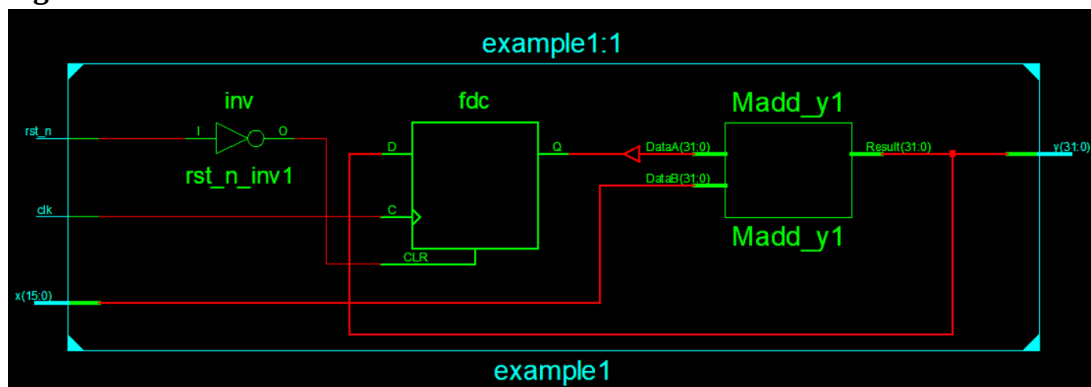
end
initial
begin
x= 0;
for(i =0; i<10; i =i+1) // generating input values every clk cycle
#20 x =x + i;
$finish;
end
always // clk generation
#10 clk = ~clk;
initial
begin
#600
$stop;
end
endmodule

```

- Waveform



- RTL Diagram



2.2. In-Lab Task#02

Design an IIR filter for $a_0=1$, $a_1=-1$, $b_0=1$, $a_2=-0.25$.

Solution:

- Verilog Code

```

module example2(input signed [15:0] x, input clk, input rst_n, output reg signed [31:0] y);

reg signed [31:0] y_reg1, y_reg2;
always @(*) // combinational logic block
y = (y_reg1) + x + (y_reg2);

```

```

always @(posedge clk or negedge rst_n) // sequential logic block
begin
if (!rst_n)
begin
y_reg1 <= 0;
y_reg2 <= 0;
end
else
begin
y_reg1 <= y;
y_reg2 <= y_reg1;
end
end

endmodule

```

- **Verilog Test Bench**

```

module example2Test;
    reg [15:0] x;
    reg clk;
    reg rst_n;
    reg i;
    wire [31:0] y;
    example2 uut (
        .x(x),
        .clk(clk),
        .rst_n(rst_n),
        .y(y)
    );
    initial begin
        clk = 0;
        #5 rst_n = 0; // resetting register before first posedge clk
        #2 rst_n = 1;
    end
    initial
    begin
        x = 0;
        for(i = 0; i < 10; i = i + 1) // generating input values every clk cycle
            #20 x = x + i;
        $finish;
    end
    always // clk generation
        #10 clk = ~clk;
    initial
    begin
        #600
        $stop;
    end
endmodule

```

- Waveform



- RTL Diagram

